

Active Inference LiveStream 006.0: "A tale of two densities: active inference"

Livestream | 1:02:55

all right i hope we are live
hello and welcome this is active
inference live stream
6.0 it is october 9th
2020. i am daniel friedman and i'm going
to be
doing a solo contextualizing episode
i guess of active inference live stream
today
on the paper tale of two densities
welcome to
team com we are an experiment in online
team communication and learning
related to active inference you can find
us on
twitter at inferenceactive at gmail
via our public key base team or via
youtube which is where you may be
watching us now
this is a recorded and an archived live
stream so please provide us with
feedback
so we can improve on different aspects
of our work
all backgrounds and perspectives are
welcome here
and as far as video etiquette goes mute
if there's noise in the background
raise your hand etc here we are
in actim stream 6.0 and the goal of this
stream
is really to set the context for the
upcoming discussion next tuesday on
6.1 that discussion is going to be
on the paper a tale of two densities

active inference is

an active inference by ramps that at all

published in 2020 online in 2019

and what this video is going to do is

provide context for some of the ideas

types of models philosophical threads

and vocabulary

of the ramstead at all paper in other

words it will help contextualize

why this topic is being discussed why

there's a regime of attention on this

topic so to speak

here are the sections of this video

first

i'm going to frame the problem as being

about what the free energy principle

and active inference say about agents

and the world and their relationship

and then to contextualize different ways

of thinking about that we'll learn about

bayesian models and inactivism just

brief brief takes of course you could go

a lot into depth on different takes here

i'm just trying to really

cut in broad strokes and then in the

last part

of the discussion today we'll talk a

little bit about the philosophy of

science

as far as the difference between a

principle a theory and a hypothesis

within the first in sphere using uh his

response to an interview question so

next time

on uh number 6.1 we're going to discuss

the paper with anyone who wants to

participate so

please save and submit your questions if

you're interested
in participating or if you're watching
it after ask your questions then or if
it's before
get in touch with us if you want to
discuss all right so here it is
the real topic that we're going to be
addressing here
is what does the free energy principle
and
active inference say about the
relationship between the world
and agents and how we think about this
formally in terms of how we model it
with computers or just
cognitively as well as how we experience
it
it's just all important so this is a
pretty general topic and so
here's one example of a world and an
agent but
world and agent are not only used
interchangeably as we'll later find out
because
one one agent's world is another world's
agent
and so on but really just because this
is a
pretty general framework of relational
modeling so
it doesn't have to be you know the the
globe or the world as you exactly think
of it and agent as in a robot agent
we're going to be probably thinking
broadly about what it means to be
connected for two things
and this video certainly is just an
introduction

to the context of some of these ideas
it's not a review
or meta-analysis or final word it's just
as i was reading through the paper it
was how i structured the information
and i was how i was uh thinking it might
make sense
to read some of the quotes and use those
as springboards to
talk about the bigger questions uh
as usual trying to start from the active
inference
perspective and then reach out to the
broader questions as well as
uh go the other way i guess and uh
in this video specifically as it relates
to free energy
principle i'll just provide the punch
the punchline first
the punch line is that generative models
are not structures
they're embodied and enacted regimes of
policy attention
and behavior so if it doesn't
make sense to you yet why that's the
punch line or why that's
one of the conclusions essentially of
this paper
then just keep listening and learning
sorry
so in this talk specifically the
structure is going to be like this
first we'll go through the goals of the
2020 tale of two densities paper
read through the abstract then we'll
talk about bayesian
inference models kind of steel manning
them putting them in a strong light

then talk about inactive and action
models pretty broadly though there's a
lot of nuance in both of these
categories then talk about agent world
modeling
in fep and active and
previewing mainly by reading quotes from
the paper
and highlighting specific claims um
highlighting
how active inference and free energy
principle build on this
uh map and territory that are in play
with bayesianism and activism and again
that underlying question
is what do we have to say about how the
world and asians and agents and agents
are related to one another
all right so the paper is called a tale
of two densities
a literature reference there active
inference is an active inference by
ramstead kirchoff and fristen in the
journal adaptive behavior in 2020
and the goal of the paper which i
believe is the first sentence of
the article is the aim of this article
is to clarify how best to interpret some
of the central constructs that
underwrite the free energy principle
fep and its corollary which means
related theory
active inference in theoretical
neuroscience and biology
namely the role that generative models
and recognition densities play
in this theory aiming to unify life and
mind

and they'll be a consistent color scheme
for the generative model
and the recognition densities because um
we're going to be looking to integrate
these two
different ways of thinking about agent
world dynamics
in the bayesian and the inactivist
worldviews
and so what we're going to talk the most
about here are
what these two densities are so you can
ask in the book what were the two
cities here we're going to ask what are
those two densities
then we can ask what is this tail that
integrates
these two densities what weaves them
together into one
narrative which helps us understand and
uh helps us learn
so let's go first to the abstract we
already read the title which is the
first way that the authors want to
represent their work now we're going to
look through the abstract
and then later on we'll end up defining
and working through a lot of the words
specifically but
starting with the abstract is the first
thing that you're going to read about
the paper
the aim of this article is to clarify
how best to interpret some of the
central constructs that underwrite the
free energy principle
and its corollary active inference in
theoretical neuroscience and biology

that may even be what we already read so
same meaning there
namely the role that generative models
and variational densities play in this
theory so those are the two
densities that we're going to be talking
about we argue
that these two constructs which are the
two densities the generative models and
the variational densities
have systematically been misrepresented
in the literature
because of the conflation between the
fep and active inference
that's why at the end we're going to be
really specific about the difference
between those two
on one hand and the distinct albeit
closely related bayesian formulations
centered on the brain
variously known as predictive processing
predictive
coding or the prediction error
minimization framework and we'll return
to
how the prediction or the predictive
brain
is uh related or different as a
from a philosophy of science perspective
and from a
computational perspective i guess all
right second part of the abstract
more specifically we examine two
contrasting interpretations of these
active inference
type models a structural uh or really
these two
variational densities the things that

we're eventually going to see brought together an active inference these okay i i believe but if it's unclear i'm not sure a structural representationalist and an active approach and that's going to be our whole main discussion we argue that the structural representationalist interpretation of generative and recognition models does not do justice to the role these constructs play in active inference under the fep so the main theme that's going to get returned to is that the computational side is very tractable and stable however it misses some fundamental components that are going to be really complementary to the insights that are brought from the inactive approach we propose we propose an inactive interpretation of active inference what might be called an active inference in active inference under the fep the generative and recognition models are best cast as realizing inference and control the self-organizing belief guided selection of action policies and do not have the properties ascribed by structural representationalists so that's just saying structural representationalists are wrong about properties in what they

think
because of actually generative models
being
basically cybernetic policy selectors
and
their claim is that that makes some of
the
stances held by structural
representationalists untenable
so let's go to the paper here's a quote
the question that shall occupy us is how
best to understand the function
and properties of the generative and
recognition models in active inference
under the fep
what does it mean for one theory to be
under the other we'll return to that at
the end
in light of the active processes
involved in orchestrating
maintaining and updating these models in
particular we examine
two contrasting interpretations of these
models a structural representationalist
that's
one of the schools of thought here and
an inactive interpretation so two
different interpretations
so we can summarize this up as a
structural representationalist versus
inactive whose hot take on systems will
reign supreme
so first let's just try to understand
what each of these
groups are really bringing to the table
because uh
that's really hopefully a
strong starting point from each of these

two sides can lead to a better synthesis
in active inference so in bayesian
inference it's a big
area this is just one way of presenting
it you can definitely watch other
lectures or learn from a lot of other
more
skilled people in this specific area
about bayesian approaches so
this is just one representation i was
trying to
make a compromise between things that
were implemented but there's so
many flavors and colors of what's
implemented in the literature
so i just wanted to start with a really
brief
overview of the bayesian inference this
is a
two level bayesian inference scheme so
variously
these two elements are known as priors
and hyper priors
or priors in the data but here on the
left side
we have data which are reflecting
sensory observations
like that of a sensor and then there's a
generative model
for example that generates sensor
readings and so
the the question that is asked in the
transformation
from the sensory observational data the
lower level model
on the left side to the generative
model of the data which is like often
the higher order parameters in the

computational model
the question is how do these
observations influence
or update the generative model of the
world and so
you can think about that as sort of like
empirical data coming in
altering your parameters that represent
deeper stances about the world so you
have a sensor that fluctuates more
rapidly
and then you're doing some sort of uh
stabilization
of the overall readings of the sensor
and that's like your higher order
parameter
just to give one of many samples and
then the other question that's being
asked
in the transformation from how the
parameters of the higher order model
influence the data is basically
what observations are emitted here given
the generative model of the world or the
generative parameters of the world which
includes
different things like world states
causes etc so here's some of the
features
and advantages of bayesian models so
first they can extend to basically
arbitrary
graphical models which doesn't just mean
models you can draw it means models that
you can graph
so it turns out that a lot of classical
and uh non-classical statistical
techniques

can be reduced down to this type of a scheme
at the highest level statistically
it's discussed in carl firsten at all's textbook related to spm
how these kind of bayesian inference models relate to various empirical data sets so potentially of interest for people who want to learn more about
that side the next thing that's good about these models
is that they can be used to model multi-scale systems so
through space and time of different scales they have very tractable computation
and implementation on hardware so you can take
a data set and basically get useful information out of it using this type of a scheme so it passes the sort of
utility check it's pretty flexible with having sparse or dense multivariate data and i'll tie that to the second point here which is that
you can be empirical in the sense that you're learning and updating your position based upon data but also you're formalizing the use of your prior information
you're neither just informally combining your prior information
with your uh newer information naively just saying oh well they're both equivalent because i have one of each one of the past one of the present

uh it provides more formalized ways of dealing with information and depending on how much you rely on your prior information you may only need very sparse inputs from your sensor for example you might only need a thermometer in a certain location to know whether it was in one climate regime or another these models can also be fit with expectation maximization algorithms that's also discussed a lot in fristen's spm textbook and uh work just and in the work of people who have done a lot more on expectation maximization algorithm um learning then i'm going to go into detail in here but it's a good topic to look into and one that i'm still learning about too uh kind of as stated in a few points these models are pretty deeply related to statistics of various kinds as well as other mathematical areas so let's just go to the structural representationalist interpretation first structure i think that there's probably wide latitude in what the structures mean and there were several citations that characterize the structural representationalist viewpoint so i'll let them speak to what structural representations and structural representationalism

is i would say structures could
represent data structures
it could represent things having
structure
in the sense that they're
distinguishable from something else so
structure could be pretty broad
representations that's a whole topic
that i understand there's
also a lot of debate over and we've read
some other
act inflated papers on representations
and uh their strengths and weaknesses
and
i i just think it is what it is and
this structural representation is
holding the views that the authors have
in their citations
and that's where we'll sort of end the
discussion on the bayesian
inference models and think about these
are all really good features of the
model
what is going to be something that's
going to complement that from a
pragmatic and philosophy of science
perspective
so the second category of models is this
inactive
or active models and here
it's a lot more focused through a
tradition
of being action oriented and ecological
it's more oriented about the relation
between the world and asia
so this comes from a lot of areas like
systems and ecology so
the two questions that are being asked

in the inactivist perspective
are first off how the world is
influencing the agent
is a question about how perception is
influencing or updating
the action tendencies of the system and
then
the way that the agent is influencing
the world or that
causal arrow is kind of asking what
actions are entertained by internal
models and entailed through control
mechanisms of the agent
so both of the questions are about the
joint
niche between the world and the agent
which will be formalized later
we'll find as a markov blanket but for
now we can just think about the world
and the agent being connected through
this perception action loop
so what are the cool parts about the
inactive
or the active perspective so it's
interdisciplinary
which encompasses a lot of interesting
work in
areas like art cybernetics ecology
psychology
dynamical systems sociology and tech and
evolution so kind of
also things that the bayesian models
have been applied
in so it's not unique to this but
some of the work in these areas is very
interesting
there's a qualitative and a relational
and experiential aspect

and anything in the computational domain
it's usually a question about efficiency
whereas things that are in the
qualitative domain often it's about
concordance with our experience or
potentially even other things
so uh it's an interesting and different
development of theory to go down the
route of more in touch with
phenomenology phenomenologists
experience per se
rather than more implementable on modern
computational hardware
it naturally relates to things like
enculturation
developmental processes sociality
because
one agent the world could be the
body of another agent or it could be a
small community or something like that
it turns out that this
inactivist perspective is more amenable
to
this externalism which we discussed in a
previous active stream
and other kind of radical embodied
cognitive
theories so i i'll just
leave that there that's kind of more of
a connector but again there's more and
more
fusion between these two approaches as
evidenced by the discussion that we're
having today
but these inactive approaches have
definitely a very
different um literature
base different citation networks that

are linking these ideas so
i'm just phrasing them in a way that's
not representing
priority or importance just sort of
throwing out different features
to differentiate it strongly from the
bayesian perspective computational
perspective
all right inactive models also
interestingly
and this is one of the reasons why
they've recently received a lot of
attention
is that they're amenable to different
kinds
of things that maybe people would call
physical computers in 2020 like ant
colonies
slime molds landslides quantum systems
all these different things that uh do
what can only be described as special
types of
algorithms or computation yet
they don't do all problems so they're
they're more like a bitcoin miner
than a uh turing complete processor and
computing language
so they're specialized computers if
they're computers personally i don't
think the computational metaphor
applies so well because i also think
that this is a
perspective that takes us beyond
computationalist metaphors
of the world sometimes
but at the very least the inactive
perspective highlighted
all these ecological dynamics of

intelligence that are only now being
appreciated from a more quantitative
perspective
and also these active models as you
might guess are part of a broader
tradition of action and pragmatism so
it's a
another interesting area that kind of
intersects there
so here's the simplification slash uh
side by side
first i'm just gonna again try to
distill it another
level down without uh
leaving too many false positives or too
many false negatives so
if someone knows better about these
areas totally i'm
open to just correcting it and changing
it i'm just
as i read through the paper i just mock
this up to
try to understand how these theories are
different and similar
so a sort of quasi-bayesian structural
representationalist view
we can distill down to this we can say
that there's the data
parameters and the hyperparameters or
just the parameters of one model and
that of another and this can be
again stacked up multi-level all these
different things that we discussed
earlier
and the way that the data like coming in
from the sensor
update the hyper parameters like the
beliefs about the world

are through what is going to be called
the recognition model by that community
on the other hand the way that the
predictions about the world state
are used to generate the data parameters
is a generative model and that is just
like

also considered sensory inference now it
turns out that there's a lot of model
and inference being thrown around like
isn't everything a model and all this
sort of stuff

yes it does um get a little confusing
and so i've tried at a few points to be
really clear and specific but sometimes
in quotes

um they uh will use like model
or who's doing inference where and we're
just going to be pretty specific about
what all the connections and the pieces
are

and then hopefully it'll be clear um
what yeah what's similar and different
between these two approaches
so the outcome of the bayesian model
like a

successfully converged model that's
running here on a computer
it represents a statistical converge
multi-level model that represents the
structures of the world

so let's just think about the data
parameters like the visual
field like the sensor data on the retina
so to speak and the hyper parameters
being like where objects are
and so even though there's no color
vision in the periphery

and even though there's low resolution
vision in the periphery still
i'm able to have a consistent uh
relationship things look pretty normal
because
the sensory data are being generated in
basic
non-surprisal with the recognition
modeled sensory data
so that would represent a sort of
bayesian computationalist
perspective on vision and error
correction
and optimization on the other hand we
can
side by side with the inactive
perspective so the inactive perspective
again situates the agent in the world as
being linked through perception and
action
and the outcome of the inactive
perspective is an embodied and
ecological
action sequence or like some sort of
niche interplay
from an embedded agent who makes or is
behavior
and this is a lot of the inspiration for
things like plants
intelligence or something like that it's
just oh well what time scale and
what problems are you asking the system
to solve and all these things
that ultimately are the decisions and
the classifications of the scientists
or so it said uh are used to talk about
the different intelligence or
perception or action capabilities of

different agents
so let's try to unite this
under what is said by free energy
principle and active inference
in this gap explanatory and causally
between the world and the agents so on
the top
of this slide there is a figure one from
the previous paper we discussed in five
i believe and several other papers
and it describes this minimal loop
between the organism
and the environment so this is the first
bridge
uh connector point so if you're on board
with just
the questions about the world and the
agent then we're still
talking about just exactly that
and we want to build on the work
and the perspective of the different
insights from a few different areas
first from the schools of action
learning about things like inactive
embodied ecological practice
socio-technical and experiential uh
aspects of the inactive view that we
discussed a few of earlier
from the more inference side again
there's a lot of directions which
inference is being used in but it's one
of the words
um in active inference and so that's why
i'm using them
here and on the more inference side
we can think about drawing the insights
from
bayesian and computational approaches

statistical approaches
modeling uh deep learning robots so all
these kind of
interesting things that think of the
world
whether uh conceptually or just from a
scientific modeling perspective from a
computational perspective
and then also there's uh another level
that as we'll see is related to the
roots of active inference in the free
energy principle
but there's this whole root layer of
physics
and specifically hear things like
invariant
sets and manifolds patterns multi-scale
dynamics collective behavior
phase transitions as well as potentially
a bridge to things like unconventional
computing or more advanced or more
tractable world modeling
so here we've gone
one level deeper into that world agent
mapping so we still have world on the
left side
in the teal and then agent on the right
side
in the darker blue except they're
labeled now as internal
and external states and they're linked
through
these two specific types of states
called sensory and action
states so i'm going to go through a list
of
just features about the fep and active
sort of skeleton of how the agent in the

world
are related to and then see if that
covers this
diagram on the bottom left we'll see so
internal and external states so that's
respectively agent in the world
states are linked through situation and
model dependent
blanket states so it's not like there's
just one
simple barrier there is a something that
is contextualized that's a
functionalized
barrier in the context of the organism
or in the context of the scientists
modeling
it and so the states can be coarse or
fine-grained
and that's really a map territory
situation
so uh depending on how
coarsely grained success in a situation
can be uh
basically achieved then even very
corresponding
of certain strategies will work for
example and one thing that we'll return
to again and again
is that active inference is about this
relationship between the sensory input
and the action selection
not necessarily just the convergence on
the hypothetical model it's about the
action
so um there's this dual instrumentalism
and i think it was a really nice
sequence of points brought up by maxwell
and others in the one of the five

discussions about this dual
instrumentalism and so
to kind of rehearse what those two
instrumentalisms are
so the first usage of these types of
partitioning models
is as a scientific framework because any
system of interest
you can identify you can identify its
inputs and its outputs and its internal
states and its external states
and you can almost think well what would
what would make this false how could i
be outside the system and model it in a
way where
there was internal states that um
influenced external states without
action
or vice versa so that might be something
that we
don't know how to measure or it could
happen but if it's a part of your
model you could specify it with a
different topology we'll return to the
question of whether you could specify
different
partitioning and have it all work out
but i'd suffice to say that for now
we're just thinking about the skeleton
of this relationship between the
different states
and as far as modeling from a scientific
framework
perspective goes you can basically
partition any
measurable system into this kind of a
model
the part that was referred to as

speculative in the previous paper
was as to whether this is actually what
organisms do
and that's sort of a philosophical area
that let's touch on
later on uh and in ongoing discussions
because i think that's sort of
one of the areas that we return to uh
almost every week in acting extreme so
just
look at this diagram think about whether
you agree with the way that states are
partitioned
and keeping in mind the multi-scale
insights from some of the previous
discussions as well and could there be
other ways that these are connected
and then we're going to want to
specifically draw together the bayesian
and inactivist ways of thinking about
action okay
so here we go to the section where i'll
read some more quotations from the paper
and just highlight draw their exact
argument
about what the different schools of
thought bring into play
so first they characterized the
structural representationalists as
having a good take on their own
models but missing action as
a fundamental consequence of
a real far from equilibrium agent
existing in the world
so the primacy of action
means that reading from the second
paragraph
we argue that the structural

representationalist interpretation
of generative and recognition models
these two bi-directional arrows
while providing an accurate description
of these constructs as they figure in
some version of bayesian cognitive
science

so in other words being a disciplinary
tool

does not do justice to the generative
models and recognition

densities that figure in active
inference under the fep

so in other words the
interpretation of how those two types of
models

agents and world and models and hyper
models how those two things are related

uh the computationalist perspective or
the the so-called structural

representationalist perspective uh

it's missing some aspect uh and then

contrasting with the second red

underlined part here

in contrast to these other bayesian i.e

purely bayesian theories

which are in effect theories of the

structure function and dynamics of the

brain

active inference is a much broader

theory of adaptive phenotypes

that centers on the control of adaptive

behavior and that emphasizes the tight

coupling and circular causality between

perception and action

so that's the part that draws it more

naturally to systems thinking

and to cybernetics

um all right so structural
representationalists argue that
generative models function
as structural representations so no
surprises there
with representational content okay
a great summary of this view which is a
nice thing to say about the authors
reads that predictive coding theory and
this is being pop
provided as a specific
kind of sub model within this bayesian
brain
structural representationalist paradigm
which we'll return to at the end as far
as their formal relationship
predictive coding postulates internal
structures so that's where you get the
structure apart
who's functioning inside of a cognitive
system closely resembles the functioning
of cartographic
maps cool it might be said that on the
proposed interpretation of the theory
cognitive systems navigate their actions
so
is it not action-oriented through the
use of a sort of causal probabilistic
maps of the world
so there is a possible uh
place for a structure so it's the maps
are the the representations that are in
the uh neural system
these maps play the role of
representations within the theory
okay believe it or not i guess it so
specifically this map like role is
played by the generative model

it is the generative models that similar
to maps constitute action guiding
detachable
structural representations that afford
representational error detection
so here's what they're proposing they're
saying that
generated maps from some sort of map
generator
function are used to then do
policy and agenda setting so action gets
separated from second layer estimation
in this way that's characterized as
detachable
in that right side of the third line
from the bottom
so it's uh that's the
perspective of the authors of this paper
characterizing
hopefully in a positive light the
structural representation lists
so those are all interesting things i
think that are cool
as well as all the aspects of these
models that are
beneficial discussed earlier so on the
other hand let's look at what inactivism
brings to the table
in this section we unpack the
implications of the pragmatist view for
understanding the relations between the
generative model
the generative process and the
recognition model according to our
pragmatist interpretation
the organism embodies the recognition
density and entails the generative model
as a control system

we then formulate a direct critique of
the claim that generative models are
structural representations
so they're going to say because of
embodied
enacted whatever it is
the generative model is a control system
and then because of the control system
and not a map of the world like where
are the structures not where the
processes of the world there's no
google maps for processes yet
but they're going to formulate a
critique in the end
that generative models are not
structural representations because they
are control systems and control systems
are not structural representations
they're something else process
realizations or something
so here's what inactivism ends up adding
it ends up providing a pretty natural
pragmatic and action-oriented
perspective on active inference modeling
it brings
all of these 4e 5e cognitive
perspectives in this is really an
interesting area of research
maybe there's other adjectives i can add
here but
it just usually is a bunch of ease
inactivism is something that natural
systems can carry out
and it's something that we can grasp at
least at the atomic level even simulate
sometimes computationally
and what is
being gained by phrasing the entire

system as a control system
for action rather than having a
so-called detachable
interface like we were seeing in the
previous approach um by framing
explicitly about action
you end up training on small or large
data sets and
evolving iterations of systems with an
eye towards function
rather than with an eye towards
structure
and that can look like a lot of
different things in different areas so a
few other
examples of previous work in the control
systems area are like cybernetics
also control systems can include
elements that are
game players they can include uh
elements of control that have to do with
swarm control as well
this was an interesting paragraph from a
technical point of view
active inference and perceptual
inference are not
merely two sides of the same coin
instead
active inference is the name of the
formulation for policy selection
what advocates of the bayesian brain
call perceptual inference is just one
moment
of the policy selection process in
active inference under the fep
namely state estimation so your your
your big deal is just a parameter in our
model

the issue we want to press here repeat
is that the active inference framework
implies that perception is a form of
action
that is action and perception cannot be
pulled apart as they sometimes are in
the bayesian brain framework
so that's that detachable critique from
the inactivist to the
um structuralism they say because you're
thinking structurally whatever that
means
you have some detachability between form
and function
we have a non-distinction between form
and function
because there's uh all these different
inactive things happening
so we're gonna have a better perspective
because we can already
computationally model our perspective
too just like you could
so closer to the end of the paper
they write in summary the role of a
generative model
is subtle in active inference the
generative model itself
never actually exists outside the
dynamics that is
outside the adaptive actions and policy
selection of the organism
well and so i'll just note there that
adaptive means that there's a striving
for
success but just because something is
adaptive doesn't mean
everyone adapts within the dynamics it
provides

a point of reference or definition of
variational free energy
or more precisely a definition of the
gradients with respect to internal and
active states
given that the vicarious realization of
the generative model through a
minimization of the variational free
energy
can only be through action and changes
in internal state
we can think of the generative model as
being enacted and of the recognition
density as being embodied
so what is being said here is
because there's sort of this
thermodynamic forwardness
that drives the active inference process
as far as far from equilibrium systems
doing what they do
because those two things are enacted and
embodied
and because it maps onto some of these
previous distinctions that were
raised and at least computationally
resolved in the computationalist
paradigm so we want to draw those
mappings about data
and about higher order perceptions of
data map that onto our understanding of
the world but
not end up computationalizing the world
per se that's one implication there's
probably others
and then they say this speaks directly
to embodied
and inactive approaches in cognitive
neuroscience

and provides a computationally tractable framework for the metaphors mobilized by these paradigms

so they're again they're saying like we'll be able to computationally so more attractively do some of the stuff that people thought philosophically or ecologically would be cool

representationalists argue that generative models encode exploitable structural information about the world

our analysis suggests that this is false indeed in this article we sought to underpin the claim that generative models do not

encode anything directly they are rather expressed in embodied activity and leverage information encoded in the recognition density which is an approximate posterior belief or best guess

assuming our conclusion is correct and why wouldn't you

our inactive inference proposal serves to free us from a standard but flawed philosophical assumption about the nature and explanatory basis of cognition

and adaptive behavior so that's the closing words of the paper

and they come out hard with the implications of action on the bayesian computationalist worldview

let's look a little bit about how active inference

steps into that gap between the agent
world gap and then also between the
bayesian computationalist and then the
more inactive
and embedded approach
so here's another quote from paper they
write
in active inference under the fep which
we'll again we'll return to
the generative and recognition models
are best cast as realizing inference and
control
the belief guided selection of action
policies so again
more like cybernetics less like image
classifiers
and do not have the properties ascribed
by structural representationalists
again read them if you want to figure
out exactly what you think they actually
are talking about
we thus provide a philosophical and
information theoretic justification
for an inactive view of generative
models under the fep
so if you read the paper
you might learn some more details about
why they argue that the generative model
is an activist in active inference
under the auspices of the fep but here's
what they end up coming to
they end up linking these two states now
i've actually
not put any labels on either of these
two nodes to emphasize
that which the there's a wide range of
relationships that the nodes can have
because it's again a general graphical

model and also it's just going to depend
on the scientists and
how it's being modeled um from a
scientific
instrumentalist point of view and
then if you're talking about organism
and niche or organism and organism like
improvisation
then again it depends on the situation
and how you're going to be
measuring it so we'll just leave the
labels off but
the model that was like the one that
goes
from the um data like the retinal data
to the higher order model model of the
world is still in orange on the top
and that's the generative model or the
inference
and then the recognition models
are um actually this one might be
flipped i'm not exactly sure
but so sorry if the colors are messed up
on this one slide
but the key aspect is that inference
is um what's happening from the data
to the higher order priors of the world
on the top in orange
and so that's like a signal processing
perspective
which lends itself naturally to
predictive processing and all these
other
things we'll get to and then on the
bottom
you have on the recognition models side
you have control so these models are not
just being used

to recognize stimuli in terms of
making structural representations of the
world or updating them
but rather in terms of their embodied
and inactive control
policies so because of this inactive
insight or interface in the pragmatic
term
the fep frames itself especially in this
multi-level
embedded way as being something that
integrates the two models by allowing
for
hopefully the mathematics to be
partitioned the way that we
expect and would hope that these kind of
well
behaved bayesian models sometimes can
behave but also
have all the richness of the inactivist
perspective
and be able to integrate these
multi-scale models
so we can think of kind of a few
generations of the active inference
development
and just think what are the
commonalities here in
the different waves of the development
and this is uh
as per a recent discussion that was
really helpful hopefully i'm
representing it roughly accurately with
uh maxwell
and axel and axe and alec and uh in the
early phrasings of active inference
there was a local
like a one step you can think

expectation maximization algorithm
so only looking one step ahead in time
led to
relatively local alignment of the model
with the sensory data
a second iteration or increase
in the nuance of active inference came
with this paper on inactive inference
and others
which introduced temporal depth and
niche into the equations
as well as things like planning and
agent embeddedness and and
the beginning surveillance and then in
some of the more
recent work like of 2019 and 2020
there's
work related to agent learning and
affect and
these sophisticated or counterfactual
aspects
of action and cognition and
counterfactuals about belief
um all the stuff that i really would
love to learn more about so
in the next and uh last
part of this stream we're gonna
talk just first coat of paint about the
connection between
a few different topics and a few
different philosophy of science topics
so the first set of topics the
scientific topics
are the bayesian brain predictive coding
active inference and the free energy
principle
and then we're gonna see how these
different

frameworks or theories or principles or
whatever they are we'll define them
how are they all linked specifically as
theories frameworks
hypotheses and connected by
falsification or other mechanisms of
comparison
and then on the right side is uh this uh
meme

and so on the top right this is sort of
reflected

uh to get the punch line first here too
on the bottom

um is a 2018 alias bulletin
interview with carl fristlin and our
departed dearly colleague martin forger
and in this interview there's sort of a
philosophy of science progression that
i'm going to walk through a little bit
of today and lay out
hopefully to clarify the differences
between some of these different theories
or what level of science are they really
striking at

and uh in this meme on the top is the
essentialist and the positivist
perspective so the essentialist being
the idea that the uh objects or things
have meaning
in themselves not just as in relation
with other objects

and positivists being the idea that
positive claims about
nature are made through measurement and
science then in the second panel is the
utilitarian so defined based upon
what they perceive their value
functionally to be and then

falsificationist

which as we'll return to is about

a theory that makes a prediction that

would make it incorrect

or make it uh untenable or increasingly

untenable to hold

and then the third panel is the

pluralist and the bayesian

and so in this world view there's

understood to be a balance between

novelty and conservation

and that a somewhat optimal balance for

certain types of

ultimately utilitarian or pluralist

views a certain value parameters can

ultimately be

struck but there's never going to be a

perfect single value because there's

still this question about values to

bring to the table so not to take it to

world modeling but

you know you're thinking about these

issues as bayesians pluralists

and then in this paper fristen just sort

of has an

another level that uh almost merits this

following chart that i'll show that

really lays out again what the fep

and the active are and how they're

related to specific testable hypotheses

because i know that's a common thing

that people

want to talk about so the green

is what martin and i had asked we wrote

in other words what is explained by the

entropic reduction within the free

energy principle that is not explained

by any model parameters in the other

frameworks so this was in an attempt to understand what a lot of people are trying to understand which is what are we getting above and beyond as a value add or conceptually or utilitarian perspective whatever it is just what are we getting by modeling this quantity as you've laid it out in these certain works and here's what Friston responded i think it is useful to make a fundamental distinction at this point that we can appeal to later the distinction is between a state and a process theory i.e the difference between a normative principle that things may or may not conform to in a process theory or hypothesis about how that principle is realized under this distinction the free energy principle stands out in stark distinction to things like predictive coding and the Bayesian brain hypothesis this is because the free energy principle is what it is a principle like Hamilton's principle of stationary action it cannot be falsified so that was obviously uh an almost inflammatory stance for many they uh fainted when they heard first then say that fap cannot be falsified it cannot be disproven in fact there's not much you can do with

it uh that's you know the other
choice sound bite right there's not much
you can do with it oh well then what are
we
doing here right unless you ask whether
measurable systems conform to the
principle
on the other hand hypotheses that the
brain performs some sort of bayesian
inference
or predictive coding are what they are
hypotheses these hypotheses may or may
not be supported by empirical evidence
so i'm going to just lay out this chart
on this continuum
to get at the distinction between all
these different philosophy of science
terms principle
theory framework everything so
this is going to be a continuum on the
bottom on the x-axis from the left side
being
more theoretical or abstract
moving through the scientific endeavor
to the statistical and the applied
so this isn't uh a ranking so much as it
is a continuum just from abstract
to uh practice based and individual
people can
can collaborate and exist across
multiple layers
of wanting to understand how this all
works
and as sort of laid out in what i'll
present as a slightly clarified and i
believe
a little bit interpreted uh
unpacking of what fristen wrote in that

paragraph

i'm gonna try to lay it out very clearly

uh

there's some gray area but i think this

is just one way to lay it out

so towards the most theoretical and

abstract

side we have principles which are

axioms and invariances so these are some

of the things that you

can't falsify and

i thought a lot a lot about that when i

was thinking about what to

say about this and one thing that it

made me think about was girdle

escherbach

the 1979 book by hofstetter

which is a really exciting exploration

about a lot of these topics related to

axioms and

logic and also

it reminded me of the law of

conservation of

stoichiometry in chemistry so not just

the law of conservation of mass or mass

energy

but of stoichiometry and chemistry

if you'll remember from taking a course

it's about balancing the equations

and so it's a rule within a certain

framework of problem solving

that equations have to be balanced now

there's also a slightly stronger more

general

belief that there's a conservation

unless there's a nuclear reaction or an

alchemical

transformation that there's a

conservation of
uh chemical elements that because
there's
such a strong conservation at the nano
level that there's also some sort of
generalized
conservation at a higher level so that's
a belief in a principle and that
ends up framing what you see how you see
it how you measure it how you compare it
it tends to be normative which means
using like a should
or an ought like oh well this motor
ought to waste heats relative to 100
efficient motor so i think we can
pretty much detach from the uh
some of the emotional psychological
baggage
on that and just think about the
scientific normativity
and then what it actually says and then
our response to it which is
absolutely valid and important is
something we can
work with and grow from the principle
doesn't need to apply to all scales so
like
the laws that say the patterns that say
that the number of molecules in a
chemical reaction are conserved
it doesn't need to say everything about
social interactions so they don't need
to be at all scales
sometimes they're sort of incorrectly
called laws i'm just not really into
thinking about
nature is having laws i think laws are
something that people make and then

there's patterns
in nature and whose last name you attach
to which law
is just wikipedia url bragging rights
what matters is the patterns that we're
discovering so again
conservation of mass energy pattern law
and this kind of returns us to this dual
instrumentalism of principles like the
free energy principle which is that
on one hand it can be about the system
like you can choose to model the system
as
neither gaining or losing energy or as
always decreasing free energy
or you could make a stronger claim that
the principle is actually how the system
is
in which case you're saying that it is
the system's functional principle that
it is under natural selection or it
does follow the second law of
thermodynamics
all right so drilling down a little bit
towards the scientific and the
statistical
a theory is a coherent set of ideas
claims in logic
and it makes specific testable
hypotheses that are drawn
from theory that's upstream and so
that's sort of the confusing thing is
from uh abstractness
uh ranking the principle is the the
ground uh
the bedrock of the theory the water
that theory swims in but it's more on
the theoretical side

so theory isn't even the most
theoretical thing
and then people say oh yes i do theory
or something like that and that can mean
a broad range of things
from anywhere from the middle to the
left side
uh theory when it's partnered with tools
and approaches
is a framework also community of
practice is important but frameworks are
important too
and the way that theories are compared
is often
through falsification or theory
comparison so falsification
is just like the best model in play is
the one that we go with
if there's some metaphor out there about
hunting ducks or something with a
shotgun i don't know
and uh there's other mechanisms of
theory comparison
theory anarchy all these wild thoughts
but um the falsificationist worldview
that like we go with what works and then
we
end up having a paradigm shift and
moving into a new set of norms around
a scientific topic like landslide either
in a little micro
sense or in a larger macro sense all
this kind of stuff
there's a lot of depth i i'm just going
to briefly just say people have thought
about these kind of theory comparators
now in the alias
interview response firstin mentioned two

kinds of theories

he mentioned state theory which he used to mean something

pretty similar to a normative principle and

those were about system states and then

he combined the normativity with it

being about the system states so it said

that there was a normative stance that

the system will

conform to a certain way of being and

it's about the states

and that was contrasted with the process

theory um which is about how

systems change which is why it's called

a process theory

all right now to get to a specific

hypothesis

a hypothesis is a specific prediction

about the system that's

under the auspices of a specific uh

theory whether it's a state theory or a

process theory

and a hypothesis is tested with

statistical

approaches it can also be explored with

visualization

so the hypothesis is tested

from a falsificationist paradigm at the

quantitative level

so at the confidence level of 0.01 or

0.001

or according to this metric those are

quantitative

sort of in the trenches falsificationism

of null hypothesis

which is that there's no influence of a

certain treatment for example

whereas the theory falsificationism
that's like one layer higher up
is a little bit more um like logical
uh battles being fought no single t-test
brings down a theory but can take down a
hypothesis and so let's
break this down for the free energy
principle specifically so
as fristen had mentioned the free energy
principle
is this principle and i think in future
times it would be cool to talk about
what makes it a principle what are the
assumptions what are the implications
what are related assumptions
and for now we'll just say that
downstream of the free energy principle
is active inference active inference is
um a process theory it's a coherent set
of ideas claims and logic
about the world it is going to be making
specifically
testable hypotheses specifically about
the structure of action
and about how agents interact with their
environment
and examples of theories that have
overlap with active inference um for
example
licensing the use of the people who are
structural representationalists
using the bayesian brain metaphor to
think about the bayesian brain as a
theory we can think about the hypotheses
that are specifically made by that
theory so for example the claims might
be something like
at the first level that the brain can be

modeled with bayesian stats and then at the second level like that the brain implements bayesian statistical algorithms in what it does in nature so that's that dual instrumentalism and then another area of theories that has overlap with active inference as well as the bayesian brain just didn't want to clutter it was predictive coding and two hypotheses in predictive coding are one would be sense data can be analyzed relative to now casted predictions now casting and the second level would be something like organismal sensory relays are predictive so again it's the dual instrumentalism where there's one level about how the scientist analyzes the world which is that yes we can use this framework to understand data even data about organisms or data about simulations or other systems and then at the second level it's actually the claim about the world and so in both of these cases their hypotheses because they are under an axiomatic framework reflected by the free energy principle that's why it keeps on saying inactive inference under the free energy principle and the active inference framework also draws on not just the bayesian brain predictive coding but just to label one more up

there
is there's this inactive area that we
talked a lot about
today and all the cool insights that
the inactive side has to bring to the
table
so maybe another time we could think
about what the
principles underlying and activism are
or are there fundamentals
it's an air i'd love to learn more about
are there
frameworks for inactive inference i know
that we've had some
participants on the active stream who
had a lot more experience and practice
with these kinds of things
so we want to see how we can make active
inference this
hub for realizing the ideas and the
people
and the projects coming together to kind
of make it real to show how these
ideas are all connected and uh yeah
see where we can kind of take these
ideas just conceptually
and from an impact perspective so
that's the end of 6.0
thanks for participating we provide
follow-up forms to live participants
it'll be pretty easy this one
any feedback suggestions or questions
would be totally welcome
stay in communication with us and
yeah the next uh live stream will be 6.1
that is going to be on october 13th
on the tuesday then we are going to have
6.2 on the 20th that's going to be a

follow-up discussion

so if you're interested in 7 a.m pacific
time

for 60 to 90 minutes on october 13th
or october 20th those are going to be
discussing this paper a tale of two
densities

then on october 27th and november
3rd both tuesdays same time we're going
to be

discussing another paper you can check
it out on our twitter if you want to
learn

more and if you're watching this after
those dates

thanks for watching please feel free to
still

leave a comment and uh yeah thanks a lot
for

watching just let us know if you have
any thoughts or questions

but have a good weekend or whenever it
is you're listening

bye